

The 6th International Workshop on Big Data Reduction (IWBDR-6)

in conjunction with IEEE BigData 2026, Phoenix, Arizona, USA

We successfully held the 1st, 2nd, 3rd, 4th, and 5th International Workshop on Big Data Reduction (<https://iwbdr.github.io/iwbdr20/>, <https://iwbdr.github.io/iwbdr21/>, <https://iwbdr.github.io/iwbdr22/>, <https://iwbdr.github.io/iwbdr23/>, and <https://apecs-lab.github.io/iwbdr25/>) in conjunction with 2020, 2021, 2022, 2023, and 2025 IEEE International Conference on Big Data, respectively.

Specifically, ❶ IWBDR-1 accepted 10 papers (9 full papers and 1 short paper), each of which gave a presentation at the workshop. We invited a speaker Dr. Bill Spatz (Program Manager from the U.S. Department of Energy (DOE) Office of Advanced Scientific Computing Research (ASCR)) to give a keynote speech at the workshop. ❷ IWBDR-2 accepted 8 papers (6 full papers and 2 short papers), each of which gave a presentation at the workshop. We invited a speaker Dr. Franck Cappello (Senior Computer Scientist from the Argonne National Laboratory) to give a keynote speech at the workshop. ❸ IWBDR-3 accepted 6 papers (2 full papers and 4 short papers), each of which gave a presentation at the workshop. We invited a speaker Dr. Feng Zhang (Associate Professor from Renmin University) to give a keynote speech at the workshop. ❹ IWBDR-4 accepted 4 full papers. ❺ IWBDR-5 accepted 10 papers (8 full papers and 2 short papers). We invited 2 speakers (Dr. Benben Liu from LSCM and Dr. Zhuoran Ji from Shandong University) for keynote speeches. We will continue these efforts this year for IWBDR-6.

Introduction to Workshop

Modern applications span high-performance computing (HPC), cloud, edge, and Internet-of-Things (IoT) platforms. Across all of them, data volumes have grown to strain storage, I/O, communication, and energy budgets at every layer of the system. Data reduction has become indispensable in this regime. It can shrink data by one or even two orders of magnitude. The downstream payoff is large: reduced memory and storage pressure, fewer I/O bottlenecks, shorter communication time, and better energy efficiency in parallel and distributed environments. The pressure is now acute. Scientific computing leads the way: exascale systems such as Frontier, Aurora, and El Capitan produce simulation outputs orders of magnitude larger than available storage, while large-scale experimental and observational facilities such as light sources, telescopes, and particle detectors stream raw measurements faster than disks and networks can ingest them. AI/ML training corpora and foundation-model checkpoints, in turn, now reach tens to hundreds of terabytes per run. At the other extreme, energy-constrained IoT and edge deployments must compress sensor streams in real time within strict power, memory, and bandwidth budgets before data reaches the cloud. Across all of these settings, runtime memory footprint and inter-node communication are non-negligible bottlenecks that data reduction can directly address.

Tackling big data reduction requires coordinated expertise across computer science, mathematics, and the application domains, together with a clear understanding of how application design, reduction algorithms, programming models, system software, and hardware interact under constraints of fidelity, performance portability, and energy efficiency. Three concrete drivers sharpen this need today. ❶ *Scientific computing* on exascale systems and large-scale experimental facilities, including light sources, telescopes, and particle detectors, produces datasets that overwhelm storage, I/O, and post-analysis pipelines, demanding error-bounded reduction with provable guarantees. ❷ *AI/ML training and inference*, including hybrid quantum-classical workloads, generate checkpoints and activations whose reduction must be co-designed with the learning semantics. ❸ *IoT and edge deployments*, from autonomous systems to smart infrastructure, push reduction onto bandwidth- and power-constrained devices for progressive, real-time streaming analytics.

Three research pillars. IWBDR-6 provides a focused venue for the data-reduction community to share results, identify new directions, and build collaborations. There are at least three significant

research topics that the community is striving to answer: ❶ *reduction ratio at scale*: whether several orders of magnitude of data reduction is possible for extreme-scale sciences; ❷ *accuracy-performance trade-off*: how reduction ratio, application-relevant accuracy, and runtime/energy cost can be jointly optimized across the Pareto frontier; and ❸ *information preservation*: solutions to effectively reduce data size while preserving the information inside the big datasets.

Relevance to IEEE BigData attendees. IWBD6 targets IEEE BigData attendees facing data-volume bottlenecks across HPC simulations, AI/ML pipelines, cloud and edge analytics, and experimental facilities. These workloads align with the conference’s themes, yet reduction-specific techniques such as error-bounded lossy compression, progressive reduction, and neural-hybrid encoders are rarely covered in depth; five prior IWBD6 editions have built precisely this cross-community venue.

Research Topics Covered in Workshop

The research topics covered by IWBD6 include but are not limited to:

- Data reduction techniques for big data issues in *high-performance computing* (HPC), *cloud computing*, *Internet-of-Things* (IoT), *edge computing*, *machine learning and deep learning*, and other big data areas:
 - Lossy and lossless compression methods
 - Approximate computation methods
 - Compressive/compressed sensing methods
 - Reduction methods for unstructured data
 - Tensor decomposition methods
 - Data deduplication methods
 - Domain/motif-specific methods, such as (un)structured meshes, particles, tensors
 - Accuracy-guarantee data reduction methods
 - Optimal design of data reduction methods
 - Progressive data reduction methods
 - Methods to provide reduced models/representations
- Mathematical methods with provable error bounds on data and quantities of interest
- Metrics and infrastructures to evaluate reduction methods and assess quality/fidelity of reduced data
- Uncertainty quantification for reduction methods/models/representations
- Benchmark applications and datasets for big data reduction
- Data analysis and visualization techniques leveraging reduced data
- Characterizing the impact of data reduction techniques on applications
- Hardware-software co-design of data reduction
- Trade-offs between accuracy & performance on emerging computing hardware and platforms
- Resource-constrained and/or time-constrained data reduction methods
- Software, tools, and programming models for managing reduced data
- Runtime systems and supports for data reduction
- Development of composable data reduction pipelines/workflows
- Automation of data reduction in scientific workflows
- Data reduction challenges and solutions in observational and experimental environments

Important Dates

The following dates are tentative and will be finalized after the proposal is accepted, aligned with the BigData 2026 workshop track.

- Oct. 1, 2026: Due date for full workshop papers submission
- Nov. 4, 2026: Notification of paper acceptance to authors
- Nov. 23, 2026: Camera-ready of accepted papers
- Dec. 14–17, 2026: IWBD-6 workshop

Steering Committee

- Sheng Di, *Argonne National Laboratory, USA*
- Xin Liang, *Oregon State University, USA*

Program Chairs

- Jiannan Tian, *Oakland University, USA*
- Daoce Wang (co-chair), *University of Nebraska Omaha, USA*

Program Committee Members

The list is sorted alphabetically by last name.

- Allison Baker, *National Center for Atmospheric Research, USA*
- Martin Burtscher, *Texas State University, USA*
- Jon Calhoun, *Clemson University, USA*
- Franck Cappello, *Argonne National Laboratory, USA*
- Jieyang Chen, *University of Oregon, USA*
- Soumya Dutta, *Los Alamos National Laboratory, USA*
- Pascal Grosset, *Los Alamos National Laboratory, USA*
- Hanqi Guo, *Ohio State University, USA*
- Pu Jiao, *The University of Texas Rio Grande Valley, USA*
- Shaomeng Li, *National Center for Atmospheric Research, USA*
- Jesus Pulido, *Los Alamos National Laboratory, USA*
- Dingwen Tao, *University of Chinese Academy of Sciences (UCAS), China*
- Xiaodong Yu, *Stevens Institute of Technology, USA*
- Zhaorui Zhang, *The Hong Kong Polytechnic University, Hong Kong, China*
- Kai Zhao, *Florida State University, USA*

Organization Expertise

The organizers are well qualified to organize the proposed IWBD-6 workshop.

Dr. Sheng Di is a computer scientist at Argonne National Laboratory and the University of Chicago. He is the leading developer of the error-bounded lossy compressor SZ (R&D 100 Award winner in 2021), which has been verified as an outstanding compressor for scientific datasets by independent researchers and users. He is also the leading developer of the lossy compression assessment tool Z-checker/QCAT,

which has been widely used to assess real-world scientific datasets. He is the lead developer of the H5Z-SZ filter, and a co-developer of the uniform compression interface library (called LibPressio). He has been working closely for years with multiple ECP application teams, such as ExaFEL (an ECP project of the exascale data analytics for free electron lasers) and ExaSky (an ECP project leading the cosmology research), so he has solid experience on how to integrate lossy compression techniques into scientific applications and workflows effectively.

Dr. Xin Liang is an associate professor with the School of Electrical Engineering and Computer Science at Oregon State University. He received his Ph.D. in Computer Science from the University of California, Riverside in 2019. His research interests lie broadly in the areas of high-performance computing, parallel and distributed systems, scientific data management, large-scale data analytics, and distributed machine learning. He has published in many highly competitive conferences and journals such as IEEE/ACM SC, ACM HPDC, ACM PPoPP, ACM ICS, IEEE IPDPS, ACM PACT, IEEE BigData, IEEE Cluster, and IEEE TPDS. He has received two Best Paper Awards from IEEE Cluster, a Best Paper Award from IEEE IPDPS, a Best Paper Award from IEEE TBD, an NSF CRII Award, an NSF EPSCoR Fellowship, an NSF CAREER Award, and the IEEE CS TCHPC Early Career Researchers Award for Excellence in High Performance Computing in 2024. He is one of the key developers of SZ and a major contributor to MGARD, two widely used data reduction software packages in the scientific computing community.

Dr. Jiannan Tian is a tenure-track assistant professor at Oakland University. He has extensive expertise in scientific computing and high-performance data reconstruction and imaging. His research spans accelerator-based lossless compression, extremely high-performance scientific lossy compression, and neural-hybrid data analytics. His team regularly publishes in prestigious HPC venues, including PPoPP, PACT, IPDPS, CLUSTER, HPDC, ICS, and SC. He also leads cyberinfrastructure development for large-scale data-processing scenarios, and one primary deliverable is pSZ/cuSZ, a multi-GPU-backend scientific compressor. Dr. Tian collaborates closely with researchers across multiple U.S. DOE national laboratories and university institutes to design the next-generation data processing and networking infrastructure, a foundation critical for advancing cyberinfrastructure in this area.

Dr. Daoce Wang is a tenure-track assistant professor at the University of Nebraska Omaha. He has expertise in designing application-driven, efficient data-reduction approaches for extreme-scale scientific simulations on HPC systems, with particular leadership in multi-resolution data reduction. He has published six first-author papers in prestigious HPC venues, including CLUSTER '21 (FFIS), HPDC '22 (TAC), TPDS '24 (TAC+), and SC '23/'24/'25 (AMRIC, MRZ, and STZ); all of his SC papers received Top Results Reproduced badges, and every project ships with full reproducible artifacts. His work on multi-resolution (AMR) data compression provides a robust foundation for IWBD-6's themes.

Organization Plan

- IWBD-6 will be organized as a **hybrid** workshop, allowing both in-person attendance in Phoenix, AZ and remote participation. The organizers will secure and manage the virtual meeting platform for remote attendees.
- We will build an official website to maintain all the information related to IWBD-6, such as submission guidelines, important dates, registration guidelines, and the program, similar to our previous IWBD workshop websites (i.e., <https://apecs-lab.github.io/iwbd25/>, <https://iwbd.github.io/iwbd23/>, <https://iwbd.github.io/iwbd22/>, <https://iwbd.github.io/iwbd21/>, and <https://iwbd.github.io/iwbd20/>).
- We will post the call-for-papers on CFP websites and email lists, such as WikiCFP and the hpc-announce mailing list, to receive as many paper submissions as possible.
- We will create a paper submission system on easychair to manage the paper submission, paper bidding, paper assignment, paper review, and notifications.
- We will invite 1 or 2 speakers to give a keynote speech at IWBD-6.

- We will ensure that IWBD-6 accepts **at least 6 regular papers** to fill a full session, per the CFP minimum. Each paper will receive at least 3 single-blind peer reviews from the Program Committee.
- We will ensure that all accepted papers have at least one registered author and that authors are present (in person or virtually) for their presentations, as required by the policy.

Invited Keynote Speakers

It has been a convention of IWBD-6 to invite a distinguished keynote speaker at every edition, drawn from national laboratories, leading research universities, and industry. Past editions have featured speakers including Dr. Bill Spotz (DOE/ASCR), Dr. Franck Cappello (Argonne National Laboratory), Dr. Feng Zhang (Renmin University), Dr. Benben Liu (LSCM), and Dr. Zhuoran Ji (Shandong University). For IWBD-6, we will continue this tradition and invite one distinguished keynote speaker; the specific invitee is to be determined and will be confirmed shortly after the proposal is accepted.